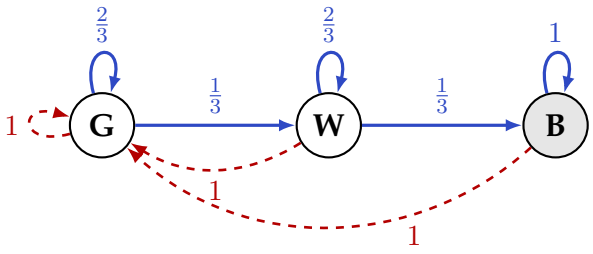
Lecture 7: Convergence Rates, Certificates, and Generalized Policy Iteration

How to use these notes. Board content in board order: figures, equations, tables, one-line statements. Today's models: Lecture 3's machine and, for two questions, Lecture 5's chain. Assumed known: Lecture 5's contraction, iteration count, stopping rule, and residual certificate; Lecture 6's policy iteration with the tie rule, the improvement theorem and its chain, "unchanged means optimal", the bound $|\mathcal{A}|^{|\mathcal{S}|}$, and the advantage table. Reading: Sutton and Barto (2018, Sections 4.4–4.7). Sources and optional reading: the last box.

Lecture 6 Left a Count

The machine again: good, worn, or broken; run or repair; $\gamma = 0.9$; it starts good.



Where each action leads. Solid: run. Dashed: repair, always back to G.

$r(s,a)$	run	repair
G	8	-10
W	4	-10
B	0	-10

This week's profit, fixed once (s,a) is.

	Value iteration (Lecture 5)	**Policy iteration (Lecture 6)**
One step costs	one lookahead per pair: $\lvert\mathcal{S}\rvert^2\lvert\mathcal{A}\rvert$	one solve, $\lvert\mathcal{S}\rvert^3$, then the same lookaheads
Steps taken, machine	78 iterations to an error of 0.01	rounds $0, 1, 2$; the stop at round 2
Steps guaranteed	$\dfrac{\log\big(R_{\max}/((1-\gamma)\varepsilon)\big)}{\log(1/\gamma)}$ iterations: 88	$\lvert\mathcal{A}\rvert^{\lvert\mathcal{S}\rvert}$ rounds: 8 here, 262,144 on the Gridworld
At the stop	certifies $\lVert V_k - V^\star\rVert_\infty \le \gamma\theta/(1-\gamma)$	ends exactly: $V^\star$ and $\pi^\star$
The run sees	the change $\lVert V_{k+1} - V_k\rVert_\infty$	the labels standing still

What Lectures 5 and 6 left. Two guarantees a world apart: 88 grows like $1/(1-\gamma)$, $|\mathcal{A}|^{|\mathcal{S}|}$ like a power of the state count. Today the second column gets the first column's count, and a certificate of its own.

Known from Lectures 3–6. For any table $V \in \mathbb{R}^{|\mathcal{S}|}$ and any policy π: $Q_V(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V(s')$, $(\mathcal{T}^\pi V)(s) = \sum_a \pi(a \mid s)\, Q_V(s,a)$, $(\mathcal{T} V)(s) = \max_a Q_V(s,a)$.

- $\mathcal{T}^\pi V \le \mathcal{T} V$, with equality iff π is greedy w.r.t. V; $V^\star \ge V^\pi$ for every π. *Lecture 4*
- Both are γ-contractions in $\|\cdot\|_\infty$; $V^\star = \mathcal{T} V^\star$ and $V^\pi = \mathcal{T}^\pi V^\pi$ are the only solutions. *Lectures 3, 5*
- Residual certificate: $\|V - V^\star\|_\infty \le \|\mathcal{T} V - V\|_\infty / (1-\gamma)$ for any table V. *Lecture 5*
- Improvement theorem: $\mathcal{T}^{\pi'} V^\pi \ge V^\pi \Rightarrow V^\pi \le \mathcal{T}^{\pi'} V^\pi \le (\mathcal{T}^{\pi'})^2 V^\pi \le \cdots \to V^{\pi'}$. *Lecture 6*
- Advantages $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$: the step changes the action where an entry is positive; no positive entry, unchanged, optimal. *Lecture 6*

How many rounds does policy iteration need? How can a run certify that it may stop before the labels stand still? And what is a policy read off approximate values worth?

Today's targets.

1. Theorem: each round gains at least an iteration, $\|V^\star - V^{\pi_{k+1}}\|_\infty \le \gamma \|V^\star - V^{\pi_k}\|_\infty$; three lines from Lecture 6's chain and Lecture 5's contraction.
2. Rounds: a loss of ε within Lecture 5's count; an optimal policy once $\gamma^k E_0$ falls below the gap Δ. Polynomial in the horizon, not $|\mathcal{A}|^{|\mathcal{S}|}$.
3. A stopping rule for policy iteration: the largest advantage δ_k certifies a loss of at most $\delta_k/(1-\gamma)$; $\delta_k = 0$ is Lecture 6's stop.
4. Evaluate less, then evaluate wrongly: the dial m, with two iterations beating the exact evaluation and five reading the worse policy; the bound $2\gamma\varepsilon/(1-\gamma)$ (stated), attained on the machine.
5. Generalized policy iteration: the loop at any grain; the map of the second half.

Theorem: Each Round Gains at Least an Iteration

Theorem (the rate of policy iteration) (Jiang, 2024, Theorem 3), (Agarwal et al., 2022, Section 1.3.2). Let $\pi_0, \pi_1, \pi_2, \ldots$ be the policies of Lecture 6's policy iteration (exact evaluation, greedy improvement). For every k,

$$\|V^\star - V^{\pi_{k+1}}\|_\infty \le \gamma \|V^\star - V^{\pi_k}\|_\infty, \qquad \text{hence} \qquad \|V^\star - V^{\pi_k}\|_\infty \le \gamma^k \|V^\star - V^{\pi_0}\|_\infty.$$

The loss of the policy falls by at least the factor an iteration of value iteration takes off the error of a table.

Why, in three lines.

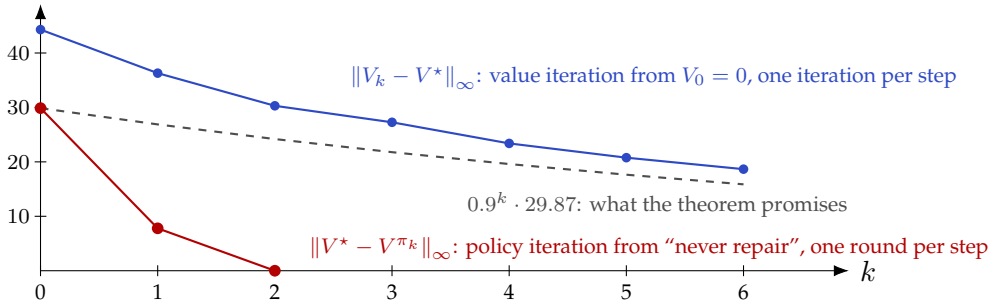
$$V^{\pi_{k+1}} \geq \mathcal{T}^{\pi_{k+1}} V^{\pi_k} = \mathcal{T} V^{\pi_k} \quad \textit{Lecture 6's chain, second term; } \pi_{k+1} \textit{ greedy w.r.t. } V^{\pi_k} \textit{ (Lecture 4)}$$

$$0 \leq V^\star - V^{\pi_{k+1}} \leq V^\star - \mathcal{T} V^{\pi_k} = \mathcal{T} V^\star - \mathcal{T} V^{\pi_k} \quad V^\star \geq V^\pi \textit{ and } V^\star = \mathcal{T} V^\star$$

$$\left\| V^\star - V^{\pi_{k+1}} \right\|_\infty \leq \left\| \mathcal{T} V^\star - \mathcal{T} V^{\pi_k} \right\|_\infty \leq \gamma \left\| V^\star - V^{\pi_k} \right\|_\infty$$

a nonnegative vector; Lecture 5's contraction □

Machine: $\mathcal{T} V^{\pi_0} = (27.5, 14.75, 14.75) \leq V^{\pi_1} = (38.46, 24.62, 24.62)$ and $\mathcal{T} V^{\pi_1} = (38.46, 26.15, 24.62) \leq V^{\pi_2} = V^\star$; losses $29.87 \to 7.79 \to 0$, ratios 0.26 and 0 against the promised 0.9.



The two runs on the machine, step by step. Value iteration loses at most a factor γ per iteration and here nearly attains it (Lecture 5); policy iteration is promised the same factor per round (dashed) and takes 0.26, then 0.

- The rate is value iteration's; only the start differs: $E_0 := \left\| V^\star - V^{\pi_0} \right\|_\infty \leq 2R_{\max}/(1-\gamma)$, 200 here, 29.87 from "never repair".
- At least one iteration per round, often many: round 0 was worth nearly 13 iterations ($0.9^{12.8} = 0.26$), round 1 all of them.
- Needs V^{π_k} exact, in the first line. What a cheaper evaluation does to the round: two sections on.

Example. Lecture 6's chain from "quit everywhere": $V^{\pi_0} = (1,1,1,1)$, $V^{\pi_1} = (1,1,1,10)$, $V^{\pi_2} = (1,1,9,10)$. Compare $\mathcal{T} V^{\pi_k}$ with $V^{\pi_{k+1}}$.
$\mathcal{T}(1,1,1,1) = (1,1,1,10)$ and $\mathcal{T}(1,1,1,10) = (1,1,9,10)$: equality at every round. There a round is exactly one iteration; losses 9, 8, 7.1, 6.29, 0.

Reading: a round is a Newton step. Policy iteration is Newton's method on $V - \mathcal{T} V = 0$ (Puterman, 1994, Section 6.4): on the piece where the greedy policy is π_{k+1} the map is $V \mapsto (I - \gamma P^{\pi_{k+1}})V - r^{\pi_{k+1}}$, and the Newton step from V^{π_k} lands on $V^{\pi_{k+1}}$, the solve. The theorem is the guaranteed part, a fixed-point step's factor γ; the handful of rounds seen in practice is Newton's fast convergence near the answer.

How Many Rounds

Corollary 1: rounds to a loss of ε.

$$\|V^\star - V^{\pi_k}\|_\infty \le \gamma^k E_0 \le \varepsilon \quad \text{once} \quad k \ge \frac{\log(E_0/\varepsilon)}{\log(1/\gamma)}$$ *the theorem k times; solve for k*

$$E_0 \le \frac{2R_{\max}}{1-\gamma} \Rightarrow k = \frac{\log\big(2R_{\max}/((1-\gamma)\varepsilon)\big)}{\log(1/\gamma)}$$ rounds always suffice

Lecture 5's count, with $2R_{\max}$ for $R_{\max}$

Machine, $\varepsilon = 0.01$: 94 rounds from $E_0 \le 200$, 76 from $E_0 = 29.87$; Lecture 5 guaranteed 88 iterations. The run: 2 rounds.

Corollary 2: an optimal policy, exactly. Let the *gap* be the smallest loss of a policy that is not optimal:

$$\Delta := \min\big\{\|V^\star - V^\pi\|_\infty : \pi \text{ deterministic, not optimal}\big\} > 0$$

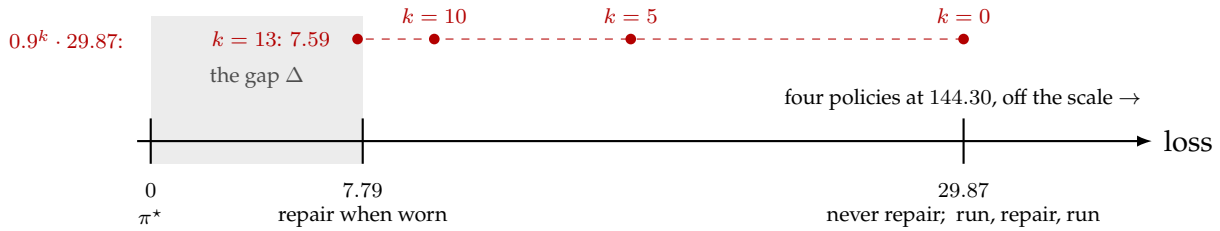
finitely many policies; each non-optimal one loses somewhere

$$\gamma^k E_0 < \Delta \Rightarrow \|V^\star - V^{\pi_k}\|_\infty < \Delta \Rightarrow \pi_k \text{ optimal}$$ *the theorem; a loss below Δ can only be 0*

so $\pi_k = \pi^\star$ for every $k > \dfrac{\log(E_0/\Delta)}{\log(1/\gamma)}$, and the run stops there at the latest

Lecture 6: the tie rule returns an optimal policy unchanged

Machine: the seven non-optimal policies lose 7.79 (repair when worn), 29.87 (two of them), 144.30 (the four that repair a good machine): $\Delta = 7.79$. $0.9^k \cdot 29.87 < 7.79$ from $k = 13$; the run had $\pi^\star$ at $k = 2$.



The eight policies' losses on one line and the theorem's envelope $\gamma^k E_0$ above it. Once the envelope enters the gap, the only loss left is 0: π_{13} is optimal whatever the path. The path took two rounds.

- Not $|\mathcal{A}|^{|\mathcal{S}|}$: about $\frac{1}{1-\gamma}\log\frac{E_0}{\Delta}$ rounds, polynomial in the horizon, logarithmic in $1/\Delta$. Machine: 13; Lecture 6 allowed 8 only because 2^3 is small.
- Δ is the model's, not the run's: near a tie (Lecture 6's $\gamma^\star$) it shrinks to 0 and the count blows up while the run still stops in a handful. Bounds free of Δ: the reading box.
- Two ingredients: the rate turned Lecture 5's count into a round count; Lecture 6's

finiteness turned "within ε" into "exactly".

γ	$1/(1-\gamma)$	iterations to an error of 0.01 taken	iterations to an error of 0.01 guaranteed	rounds to a loss of 0.01 taken	rounds to a loss of 0.01 guaranteed	round k with π_k optimal taken	round k with π_k optimal guaranteed (gap)
0.5	2	8	11	0	12	0	0
0.9	10	78	88	2	94	2	13
0.99	100	1047	1146	2	1215	2	262
0.999	1000	12,819	13,809	2	14,502	2	2828

Growth with the horizon, by computer: value iteration from $V_0 = 0$, policy iteration from "never repair". Guaranteed iterations and rounds to 0.01 use $R_{\max}/(1-\gamma)$ and $2R_{\max}/(1-\gamma)$; the gap column uses the true E_0 and Δ (7.79, 26.26, 219.03). Every guarantee grows like $1/(1-\gamma)$; the machine's rounds do not move.

Reading: round counts free of the gap. Ye (2011): at most $|\mathcal{S}|^2|\mathcal{A}|\log\big(|\mathcal{S}|^2/(1-\gamma)\big)/(1-\gamma)$ rounds from any start, as stated by Agarwal et al. (2022, Section 1.3.2). Scherrer (2016), in one idea: after $k^\star = \lceil \log(1/(1-\gamma))/\log(1/\gamma) \rceil + 1$ rounds the theorem has cut the loss by more than $1-\gamma$, which forces some non-optimal action of π_0 out for good; there are at most $|\mathcal{S}|(|\mathcal{A}|-1)$ such pairs, so at most $|\mathcal{S}|(|\mathcal{A}|-1)\,k^\star$ rounds in all. Machine: 3 pairs, $k^\star = 23$, at most 69 rounds, whatever the start and whatever Δ. Value iteration has no bound of this kind: the iterations it needs before its greedy policy settles on $\pi^\star$ can exceed any polynomial in $|\mathcal{S}|$, $|\mathcal{A}|$, $1/(1-\gamma)$ (Feinberg et al., 2014).

A Stopping Rule for Policy Iteration

The run cannot see the loss. It can see Lecture 6's advantage table.

The residual of a policy.

$$\delta_k \;:=\; \|\mathcal{T}V^{\pi_k} - V^{\pi_k}\|_\infty \;=\; \max_s \max_a A^{\pi_k}(s,a) \;\geq\; 0$$

$(\mathcal{T}V^\pi)(s) - V^\pi(s) = \max_a Q^\pi(s,a) - V^\pi(s)$; the entry at $\pi(s)$ is 0

The largest entry of the advantage table: the improvement step has already computed it. Machine: $14.75,\ 1.54,\ 0$.

What it certifies, in three lines.

$$V^\star - V^{\pi_k} = (\mathcal{T}V^\star - \mathcal{T}V^{\pi_k}) + (\mathcal{T}V^{\pi_k} - V^{\pi_k}) \qquad \textit{add and subtract } \mathcal{T}V^{\pi_k};\ V^\star = \mathcal{T}V^\star$$

$$\|V^\star - V^{\pi_k}\|_\infty \le \gamma \|V^\star - V^{\pi_k}\|_\infty + \delta_k \qquad \textit{Lecture 5's contraction; the definition of } \delta_k$$

$$\|V^\star - V^{\pi_k}\|_\infty \le \frac{\delta_k}{1-\gamma}, \qquad \|V^\star - V^{\pi_{k+1}}\|_\infty \le \frac{\gamma\,\delta_k}{1-\gamma} \qquad \textit{solve for the norm; then the theorem}$$

Lecture 5's residual certificate at $V = V^{\pi_k}$, where the error of the table *is* the loss of the policy.

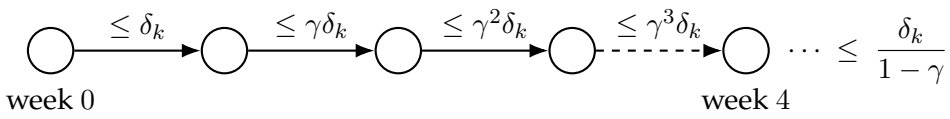
Stopping rule for policy iteration. Stop at the first round with

$$\delta_k \le (1-\gamma)\,\varepsilon, \qquad \text{and return } \pi_k : \quad \|V^\star - V^{\pi_k}\|_\infty \le \varepsilon.$$

$\varepsilon = 0$: $\delta_k = 0$ iff no advantage is positive iff $\pi_{k+1} = \pi_k$ under the tie rule. Lecture 6's stop is this rule at $\varepsilon = 0$.

k	π_k	δ_k	certified for π_k	true loss of π_k	certified for π_{k+1}
0	never repair	14.75	147.5	29.87	132.75
1	repair when worn	1.54	15.38	7.79	13.85
2	repair when broken	0	0	0	—

The rule on the machine: certified $\delta_k/(1-\gamma)$ for π_k and $\gamma\delta_k/(1-\gamma)$ for π_{k+1}, against the truth; 4.9 and 2.0 times the loss. Lecture 5's chain from "quit everywhere": $\delta_k = 9,\ 8,\ 7.1,\ 6.29,\ 0$, certified $90,\ 80,\ 71,\ 62.9,\ 0$, true loss $= \delta_k$, exactly 10 times. Both runs jump to $\delta_k = 0$ rather than fading, so for any $\varepsilon < 15.38$ the rule stops where Lecture 6's did.



What the certificate charges: the largest advantage forgone every week, forever, discounted. The chain forgoes its one advantage once and the episode ends, so the certificate is ten times the loss; the machine under $\pi^\star$ keeps running, so the two are closer.

- The rule earns its keep when δ_k never reaches 0: the dial (next section) and Lecture 8's estimated model. Exact policy iteration on a finite MDP stops by itself.
- Corollary 1 becomes computable at round 0: $E_0 \le \delta_0/(1-\gamma) = 147.5$, so 92 rounds certify 0.01 (94 blind, 76 with the true E_0).
- Lecture 5's certificate was nearly exact on the machine, this one is off by 2 to 10: what a certificate charges is a worst case over every model with this γ.

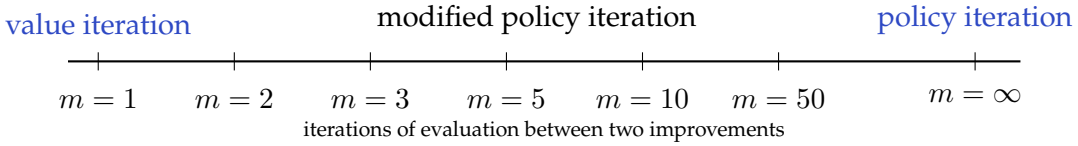
Example. Round 1 on the machine: the advantage table has one positive entry, 1.54 at W. What does the run know about π_1, about π_2, and what is true?
π_1 within $1.54/0.1 = 15.38$ of $V^\star$ (true 7.79); π_2 within $0.9 \cdot 15.38 = 13.85$ (true 0). That π_2 is optimal is known only at round 2, when the table has no positive entry.

Evaluate Less: The Dial

Every line so far needed V^{π_k} exact. Lecture 5's run never was, and still read off $\pi^\star$ from $k = 2$: how much evaluation does a round need?

Modified policy iteration (Puterman and Shin, 1978), (Puterman, 1994, Section 6.5)

- Input: the model (P, r, γ); $m \in \{1, 2, 3, \ldots\}$ or $m = \infty$; a tolerance θ; V_0 (here 0).
- For $k = 0, 1, 2, \ldots$:
  - *Improve:* $\pi_k(s) \leftarrow \operatorname{argmax}_a Q_{V_k}(s, a)$ for every s, keeping $\pi_{k-1}(s)$ when it is among the maximizers.
  - *Evaluate, m iterations:* $V_{k+1} \leftarrow (\mathcal{T}^{\pi_k})^m V_k$, m iterations of Lecture 3's $V \leftarrow r^{\pi_k} + \gamma P^{\pi_k} V$ from V_k; for $m = \infty$ the solve, $V_{k+1} \leftarrow V^{\pi_k}$.
- Stop when $\|V_{k+1} - V_k\|_\infty \le \theta$, or for $m = \infty$ when $\pi_k = \pi_{k-1}$; return π_k.



One dial. Every setting improves at every state and then evaluates the new policy for a while; only how long differs.

The two ends of the dial.

$$m = 1: \quad V_{k+1} = \mathcal{T}^{\pi_k} V_k = \mathcal{T} V_k \qquad \pi_k \text{ greedy w.r.t. } V_k \text{ (Lecture 4): value iteration}$$

$$m = \infty: \quad V_{k+1} = V^{\pi_k}, \quad \pi_{k+1} = \text{greedy}(V^{\pi_k}) \qquad \text{policy iteration, started from } \pi_0 = \text{greedy}(V_0)$$

An iteration of value iteration is an improvement followed by one iteration of evaluation (Sutton and Barto, 2018, Section 4.4). Every m converges (Puterman, 1994, Section 6.5): the reading box below.

Example. Read Lecture 5's $V_1 = (8, 4, 0) \to V_2 \to V_3(\mathrm{B})$ as improve, then evaluate once. $\pi_1 = \text{greedy}(V_1) =$ never repair (repair column -2.8); $\mathcal{T}^{\pi_1} V_1 = (14,\ 6.4,\ 0) = V_2$. $\pi_2 = \text{greedy}(V_2) =$ repair when broken (repair column 2.6); at B, $-10 + 0.9 \cdot 14 = 2.6 = V_3(\mathrm{B})$.

From $V_0 = 0$, $\pi_0 = \text{greedy}(0)$ = "never repair" (repair column -10). Round 0 for several settings of the dial:

m	$V_1 = (\mathcal{T}^{\pi_0})^m V_0$	repair, any s	run at W	$\pi_1 = \text{greedy}(V_1)$	$V^{\pi_1}(\text{G})$
1	$(8,\ 4,\ 0)$	-2.8	6.4	never repair	27.5
2	$(14,\ 6.4,\ 0)$	2.6	7.84	repair when broken	44.30
3	$(18.32,\ 7.84,\ 0)$	6.49	8.70	repair when broken	44.30
4	$(21.34,\ 8.70,\ 0)$	9.21	9.22	repair when broken	44.30
5	$(23.42,\ 9.22,\ 0)$	11.08	9.53	repair when worn	38.46
10	$(27.03,\ 9.94,\ 0)$	14.33	9.96	repair when worn	38.46
∞	$(27.5,\ 10,\ 0)$	14.75	10	repair when worn	38.46

Round 0 from $V_0 = 0$ for each m: the evaluation is Lecture 3's iterates of "never repair". Repair pays $-10 + 0.9\,V_1(\text{G})$; run at W pays $4 + 0.9\big(\frac{2}{3}V_1(\text{W}) + \frac{1}{3}V_1(\text{B})\big)$; at G run wins throughout, at B repair wins from $m = 2$ on.

Example. Two iterations of evaluation, then the exact evaluation: π_1 in each case, and its worth?
$m = 2$: $V_1 = (14,\ 6.4,\ 0)$; repair 2.6; W run $7.84 > 2.6$, B repair $2.6 > 0$: π_1 = repair when broken $= \pi^\star$, worth 44.30.
$m = \infty$: $V_1 = (27.5,\ 10,\ 0)$; repair 14.75; W repair $14.75 > 10$: π_1 = repair when worn, worth 38.46. The exact evaluation of π_0 gave the worse π_1.

- Decided at W: the repair column climbs with $V_1(\text{G})$, 8 toward 27.5; the run entry with $V_1(\text{W})$, 4 toward 10; they cross between $m = 4$ and $m = 5$.
- Lecture 6's theorem at $m = \infty$: $V^{\pi_1} \geq V^{\pi_0}$, and it holds ($38.46 > 27.5$). It ranks π_1 against π_0 only, never one π_1 against another.
- For finite m it says nothing: V_1 is not V^{π_0}. That $m = 2, 3, 4$ landed on $\pi^\star$ is this machine's luck.
- Whole runs to 0.01, by computer: $m = 1, 2, 3$ take 78 iterations ($78,\ 39,\ 26$ rounds); $m = 5, 10, 50$ take $80,\ 90,\ 200$ through "repair when worn"; the solve, 3 rounds. Stopped after the $m = 5$ round, the policy loses 5.84 at G.

Reading: why every m converges, from a V_0 with $\mathcal{T}V_0 \geq V_0$ (here $\mathcal{T}0 = (8, 4, 0) \geq 0$). Write $V_k^{\mathrm{VI}} = \mathcal{T}^k V_0$ for Lecture 5's iterates.

$$V_k \leq \mathcal{T}V_k = \mathcal{T}^{\pi_k}V_k \leq (\mathcal{T}^{\pi_k})^2 V_k \leq \cdots \leq (\mathcal{T}^{\pi_k})^m V_k = V_{k+1} \leq V^{\pi_k} \leq V^\star$$

Lecture 6's chain from V_k, then its limit

$$V_{k+1} = (\mathcal{T}^{\pi_k})^m V_k \leq (\mathcal{T}^{\pi_k})^{m+1} V_k = \mathcal{T}^{\pi_k} V_{k+1} \leq \mathcal{T}V_{k+1}$$

one more link; $\mathcal{T}^\pi \leq \mathcal{T}$: the first line holds at $k+1$

$$V_{k+1} \geq \mathcal{T}V_k \geq \mathcal{T}V_k^{\mathrm{VI}} = V_{k+1}^{\mathrm{VI}}$$

the first line; $\mathcal{T}$ monotone and $V_k \geq V_k^{\mathrm{VI}}$, by induction

$$V_k^{\mathrm{VI}} \leq V_k \leq V^\star \quad \Rightarrow \quad \|V^\star - V_k\|_\infty \leq \left\|V^\star - V_k^{\mathrm{VI}}\right\|_\infty \leq \gamma^k \|V^\star - V_0\|_\infty$$

no m is slower per round than $m = 1$

Machine: every V_1 in the table lies between $(8, 4, 0)$ and $(27.5, 10, 0)$; the rounds to 0.01 fall from 78 as m grows. The per-round rate of the theorem, however, is $m = \infty$'s alone: the tables are not V^{π_k}.

What a Policy Read Off Any Table Is Worth

The $m = 5$ round read the worse policy off an approximate table, and Lecture 5's stop hands over greedy w.r.t. V_k: what is a policy read off an approximate table worth?

Theorem (stated): the loss of a greedy policy (Singh and Yee, 1994). If $\|V - V^\star\|_\infty \leq \varepsilon$ and π is greedy with respect to V, then

$$V^\pi(s) \geq V^\star(s) - \frac{2\gamma\,\varepsilon}{1-\gamma} \qquad \text{for every } s.$$

The reason, in three lines (the proof is a recursion on the third):

$$\left|Q_V(s,a) - Q^\star(s,a)\right| = \gamma \left|\sum_{s'} P(s' \mid s,a)\big(V(s') - V^\star(s')\big)\right| \leq \gamma\varepsilon$$

an average of V's errors, times γ

$$Q^\star\big(s,\pi(s)\big) \geq Q_V\big(s,\pi(s)\big) - \gamma\varepsilon \geq Q_V\big(s,\pi^\star(s)\big) - \gamma\varepsilon \geq Q^\star\big(s,\pi^\star(s)\big) - 2\gamma\varepsilon$$

the line above; π greedy w.r.t. V; the line above again

one week's choice costs at most $2\gamma\varepsilon$; forever: $2\gamma\varepsilon\,(1 + \gamma + \gamma^2 + \cdots) = \dfrac{2\gamma\varepsilon}{1-\gamma}$

π may repeat the mistake every week

Example (the bound is attained). $V = V^\star + 1.66\,(-1,\ 0,\ +1)$, so $\|V - V^\star\|_\infty = 1.66$: the greedy action at B?
run $0.9\,(29.87 + 1.66) = 28.38$, repair $-10 + 0.9\,(44.30 - 1.66) = 28.38$: a tie, at $\varepsilon = 2.99/1.8$ exactly. G, W: run, as at $V^\star$.
Broken toward run: never repair, worth $(27.5,\ 10,\ 0)$, loss $29.87 = 18{\cdot}1.66$, the bound with equality (Williams and Baird, 1993). The wrong action at B forgoes $(1-\gamma)V^\star(\mathrm{B}) = 2.99$ every week.

The same certificate through the residual (Brunskill, 2026, Assignment 1, Problem 3). For any table V with $\|\mathcal{T}V - V\|_\infty \le \theta$ and π greedy w.r.t. V, so that $\mathcal{T}^\pi V = \mathcal{T}V$:

$$\|V - V^\star\|_\infty \le \frac{\theta}{1-\gamma}, \qquad \|V - V^\pi\|_\infty \le \frac{\|\mathcal{T}^\pi V - V\|_\infty}{1-\gamma} = \frac{\theta}{1-\gamma}$$

Lecture 5's certificate, for $\mathcal{T}$ and for $\mathcal{T}^\pi$

$$V^\star - V^\pi = (\mathcal{T}V^\star - \mathcal{T}V) + (\mathcal{T}^\pi V - \mathcal{T}^\pi V^\pi) \ \le\ \gamma\,\|V^\star - V\|_\infty + \gamma\,\|V - V^\pi\|_\infty \ \le\ \frac{2\gamma\theta}{1-\gamma}$$

$\mathcal{T}V = \mathcal{T}^\pi V$; two contractions; the line above twice

$18\,\theta$ at $\gamma = 0.9$. Through the error instead, Lecture 5's rule gives $\varepsilon = \gamma\theta/(1-\gamma)$ and the theorem $2\gamma^2\theta/(1-\gamma)^2 = 162\,\theta$: the factor $1/(1-\gamma)$ paid twice.

The table	**Its residual** $\|\mathcal{T}V - V\|_\infty$	**Certifies the table**	**Certifies the policy**
any V	θ	$\|V - V^\star\|_\infty \le \theta/(1-\gamma)$ (Lecture 5)	greedy(V) loses at most $2\gamma\theta/(1-\gamma) = 18\,\theta$
V_k at value iteration's stop	the change, $\le \theta$	the same; V_{k+1} within $\gamma\theta/(1-\gamma)$	the policy read off V_k: $18\,\theta$; read off V_{k+1} through its error: $162\,\theta$
V^{π_k} of policy iteration	δ_k, the largest advantage	$\|V^{\pi_k} - V^\star\|_\infty \le \delta_k/(1-\gamma)$	the same number: the table's error is π_k's loss; π_{k+1} within $\gamma\delta_k/(1-\gamma)$

One residual, three readings. Certifying a loss of 0.01 *on the machine:* 85 *iterations through the residual,* 105 *through the error; policy iteration certifies* 0 *at round* 2.

k	$\varepsilon = \|V_k - V^\star\|_\infty$	greedy w.r.t. V_k	true loss $\|V^\star - V^\pi\|_\infty$	bound 18ε
0	44.30	never repair	29.87	797
2	30.30	repair when broken	0	545
40	0.52	repair when broken	0	9.3
78	0.0095	repair when broken	0	0.17

The bound against the truth on Lecture 5's iterates: useless early (larger than any value in the problem), loose late (the policy has been $\pi^\star$ since $k = 2$).

- Certificates are worst cases over every model with this γ: at $\theta = 0.001$ the run knows 0.018 (or 0.162), not 0.
- Why $2\gamma\varepsilon$: greedy sees comparisons only; $Q_{V+c} = Q_V + \gamma c$, a constant shift is free. An error that differs between states, $+\gamma\varepsilon$ on one entry and $-\gamma\varepsilon$ on the other, flips one: the Example spends that at B.
- In Q form: $\|Q - Q^\star\|_\infty \le \varepsilon$ and $\pi = \text{greedy}(Q)$ lose at most $2\varepsilon/(1-\gamma)$, the lookahead being inside Q (Agarwal et al., 2022, Lemma 1.11).
- One-sided errors ($V \ge V^\star$ everywhere, or $V \le V^\star$) cost half, $\gamma\varepsilon/(1-\gamma) = 9\varepsilon$: one swap in the second line is free. Attained at $V^\star + (0, 0, 3.32)$.

Generalized Policy Iteration

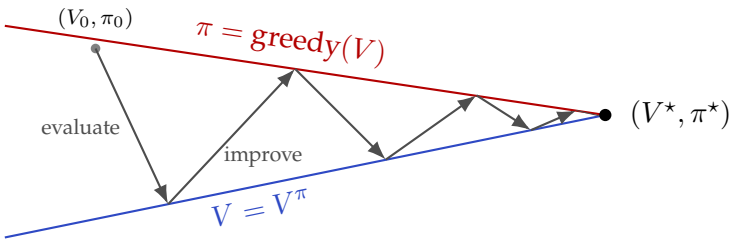
Value iteration, policy iteration, and every setting of the dial are one loop.

Two processes, one loop (Sutton and Barto, 2018, Section 4.6).

- *Evaluation* moves the table toward the policy's values, $V \to V^\pi$: a solve; m iterations of $\mathcal{T}^\pi$; one iteration; later, averages of sampled returns.
- *Improvement* moves the policy toward greedy w.r.t. the table, $\pi \to \text{greedy}(V)$: at every state or at some; exactly or nearly (ε-greedy, below).
- Generalized policy iteration: any interleaving of the two that keeps updating every state.

$$V = V^\pi \text{ and } \pi = \text{greedy}(V) \;\Rightarrow\; V^\pi = \mathcal{T}^\pi V^\pi = \mathcal{T} V^\pi \;\Rightarrow\; (V, \pi) = (V^\star, \pi^\star)$$

Lecture 6's "unchanged means optimal": both stand still only there



Sutton and Barto's picture of Section 4.6, redrawn. Each process reaches its own line and leaves the other; the only point on both is $(V^\star, \pi^\star)$. Policy iteration takes full steps to each line; a finite m takes partial ones toward the blue line.

Improvement need not be greedy: ε-greedy improves too (Sutton and Barto, 2018, Section 5.4), (Brunskill, 2026, Lecture 4). Let π give every action probability at least $\varepsilon/|\mathcal{A}|$ (ε-soft), and let π' be ε-greedy w.r.t. Q^π: the greedy action with probability $1-\varepsilon+\varepsilon/|\mathcal{A}|$, each other action with $\varepsilon/|\mathcal{A}|$.

$$(\mathcal{T}^{\pi'}V^\pi)(s) = \frac{\varepsilon}{|\mathcal{A}|}\sum_a Q^\pi(s,a) + (1-\varepsilon)\max_a Q^\pi(s,a) \;\geq\; \sum_a \pi(a \mid s)\, Q^\pi(s,a) = V^\pi(s)$$

the weights $\pi(a \mid s) - \varepsilon/|\mathcal{A}| \geq 0$ sum to $1-\varepsilon$; a max beats their average

$$\Rightarrow\; V^{\pi'} \;\geq\; V^\pi$$

Lecture 6's proof, unchanged for a stochastic π': it needs only $P^{\pi'} \geq 0$

Machine, $\varepsilon = 0.1$, from "never repair" softened (run w.p. 0.95): $(29.12,\ 14.43,\ 5.59) \to (33.30,\ 20.07,\ 19.86) \to (39.82,\ 28.87,\ 25.70)$, then unchanged: Lecture 6's path, softened, ending 4.49 below $V^\star$ at G, the price of exploring.

With the model	Evaluation of π	Improvement
Policy iteration	exact: one solve	greedy, every state
Value iteration	one iteration of $\mathcal{T}^\pi$	greedy, every state
Modified policy iteration	m iterations of $\mathcal{T}^\pi$	greedy, every state
Asynchronous DP (Sutton and Barto, 2018, Section 4.5)	one state at a time, in any order that keeps visiting every state	greedy at the states updated

The loop with the model: four settings of the same two steps, every one of them convergent.

Reading: what the model's absence costs. Without P and r, Q^π is estimated from sampled transitions: the evaluation is noisy, so Lecture 6's theorem is no longer a guarantee, and the dial's $m = 5$ round, which read the worse policy off an approximate table, is the standing warning; a deterministic policy learns nothing about its other actions, so the improvement stays ε-greedy; the two processes run one transition at a time, the policy changing under the evaluation. Convergence is proved for some of these methods and only hoped for in others: the loop is the map, not the guarantee (Sutton and Barto, 2018, Section 4.8). Lecture 8 keeps the model instead: estimate P and r, then run policy iteration on the estimate.

Without the model	**Evaluation of π**	**Improvement**	**Data**
Monte Carlo control	average the sampled returns that followed (s, a)	ε-greedy: mostly greedy, every action kept alive	whole episodes
SARSA	one step toward $R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})$ per transition	ε-greedy at every step	transitions
Q-learning	one step toward $R_{t+1} + \gamma \max_a Q(S_{t+1}, a)$: value iteration's update, one sampled pair at a time	greedy off the table; ε-greedy to collect data	transitions
Actor–critic	a critic tracks V^π or Q^π	a gradient step on a parameterized π toward higher Q	trajectories
Search (AlphaZero)	self-play returns fit a value network	a search tree sharpens the move distribution	a simulator

The loop without the model: each row is an evaluation and an improvement with samples in place of P and r. The rows are named by what they are, all to come.

Example (Jiang, 2026, control slides). Control always has a max somewhere. Q-learning's is explicit, $\max_a Q(S_{t+1}, a)$. Where is SARSA's?
In the choice of A_{t+1}: ε-greedy w.r.t. Q, so $Q(S_{t+1}, A_{t+1})$ is the max with probability $1 - \varepsilon + \varepsilon/|\mathcal{A}|$. Improvement at every step, evaluating a policy that keeps changing.

The Planning Unit, Closed

	Brute force (Lecture 4)	**Guess and test (Lecture 4)**	**Value iteration (Lecture 5)**	**Policy iteration (Lecture 6)**
Computes	V^π for every table π	V^π for one guess, then $\mathcal{T}V^\pi$	$V_k = \mathcal{T}^k V_0$, no policy fixed	V^{π_k} for a chain of guesses, each the greedy step from the last
On the machine	8 solves	1 solve and 6 lookaheads per guess	78 iterations, 468 lookaheads	3 solves and 18 lookaheads
Steps guaranteed	$\lvert\mathcal{A}\rvert^{\lvert\mathcal{S}\rvert}$ solves	one solve; no guarantee the guess passes	$\frac{1}{1-\gamma}\log\frac{R_{\max}}{(1-\gamma)\varepsilon}$ iterations	the same count to a loss of ε; $\frac{1}{1-\gamma}\log\frac{E_0}{\Delta}$ rounds to $\pi^\star$
Answer	exact	exact if the guess passes, else a failed guess	within $\gamma\theta/(1-\gamma)$; a policy within $18\,\theta$	exact; a loss within $\delta_k/(1-\gamma)$ at every round

Example. Which would you run? (a) A million states, each row of P with a handful of nonzero entries. (b) Twenty states, $\gamma = 0.999$. (c) The exact optimal policy is required. (d) P and r unknown; only sampled transitions. Answers: the table.

Case	**Run**	**Why**
(a) 10^6 states, sparse P	value iteration, or the dial at a small m	a solve is $\lvert\mathcal{S}\rvert^3$, out of reach; an iteration is one lookahead per pair, cheap when P is sparse
(b) 20 states, $\gamma = 0.999$	policy iteration	the guarantees grow alike, but a round is worth at least an iteration and usually many: $12{,}819$ iterations against 2 rounds on the machine
(c) exact $\pi^\star$ required	policy iteration	it ends exactly, within $\frac{1}{1-\gamma}\log\frac{E_0}{\Delta}$ rounds; value iteration certifies within
(d) no model, only samples	neither as written	the loop with samples for the model: the table above, and the rest of the course

Summary

Lecture 7 on one board

- Theorem: $V^{\pi_{k+1}} \geq \mathcal{T}V^{\pi_k}$, so $\|V^\star - V^{\pi_{k+1}}\|_\infty \leq \gamma \|V^\star - V^{\pi_k}\|_\infty$: each round gains at least an iteration. Machine 0.26, then 0; chain exactly one iteration per round.
- Rounds: a loss of ε within $\log(E_0/\varepsilon)/\log(1/\gamma)$, $E_0 \leq 2R_{\max}/(1-\gamma)$ (94 against Lecture 5's 88); $\pi^\star$ exactly once $\gamma^k E_0 < \Delta$ (13; the run took 2). Polynomial in $1/(1-\gamma)$, not $|\mathcal{A}|^{|\mathcal{S}|}$.
- Stopping rule: $\delta_k = \|\mathcal{T}V^{\pi_k} - V^{\pi_k}\|_\infty$, the largest advantage, certifies $\|V^\star - V^{\pi_k}\|_\infty \leq \delta_k/(1-\gamma)$; stop at $\delta_k \leq (1-\gamma)\varepsilon$; $\varepsilon = 0$ is Lecture 6's stop. Machine $147.5,\ 15.38,\ 0$ against $29.87,\ 7.79,\ 0$.
- The dial: $\pi_k = \text{greedy}(V_k)$, $V_{k+1} = (\mathcal{T}^{\pi_k})^m V_k$; $m = 1$ value iteration, $m = \infty$ policy iteration. Round 0 from zero: $m = 2$ gives $\pi^\star$, $m = \infty$ "repair when worn"; the theorem ranks π_1 against π_0 only.
- Bound (stated): $\|V - V^\star\|_\infty \leq \varepsilon$ gives a loss of at most $2\gamma\varepsilon/(1-\gamma)$; attained at $\varepsilon = 1.66$ (greedy sees comparisons, not distances); through the residual, $18\,\theta$. One residual certifies any table, value iteration's stop, and policy iteration's round.
- Generalized policy iteration: $V \to V^\pi$, $\pi \to \text{greedy}(V)$, any interleaving, greedy or ε-greedy; both still only at $(V^\star, \pi^\star)$. Every method to come is the loop with samples for the model.

Sources and optional reading. The rate of policy iteration is classical (Puterman, 1994, Section 6.4); the three-line derivation follows Jiang (2024, Theorem 3) and Agarwal et al. (2022, Section 1.3.2), written on V instead of Q, and the round count to a loss of ε is their corollary. The gap argument is folklore; the counts free of the gap are Ye (2011) and Scherrer (2016), and value iteration's lack of one is Feinberg et al. (2014). The stopping rule is Lecture 5's residual certificate (Williams and Baird, 1993) applied to V^{π_k}, with Jiang (2024, Section 2.2)'s Bellman-error check as the model. Modified policy iteration is Puterman and Shin's (1978); the sandwich follows Puterman (1994, Section 6.5), and value iteration as one iteration of evaluation per improvement is Sutton and Barto (2018, Section 4.4). The greedy-policy bound is Singh and Yee's (1994), stated as in Sun (2025, policy-iteration slides), Jiang (2026, February 12 blackboard notes), and Agarwal et al. (2022, Lemma 1.11); its tightness is Williams and Baird's (1993); the residual and one-sided forms are Brunskill (2026, Assignment 1, Problem 3)'s. Generalized policy iteration is Sutton and Barto (2018, Section 4.6), its two-lines picture redrawn; ε-greedy improvement is Sutton and Barto (2018, Section 5.4) and Brunskill (2026, Lecture 4); the map to sample-based methods follows Jiang (2026, control slides) and Brunskill (2026, Lecture 4); the cost comparison follows Agarwal et al. (2022, Section 1.3) and Brunskill (2026, Lecture 2). Optional: Puterman (1994, Sections 6.4–6.5) for policy iteration as Newton's method and modified policy iteration with every proof; Jiang (2024, Section 2.2) for the bound proved in four lines.

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